Strategy Framework (S1)

Porter's Five Forces: Rivalry (many competitors, low diff., high fixed costs); Entrants (capital, scale, patents, regulation); Substitutes & switching costs; Buyer power (concentrated, undiff.); Supplier power (few, high switching costs). Strong forces ⇒ lower margins.
Economic Attributes: Demand (price sensitivity, cyclicality, seasonality); Supply (diff., barriers); Production (capital vs. labor intensity); Marketing (B2B/B2C, push/pull); Assets (short vs. long-term).
▷ *Strategy sets the benchmark for expected ratios. Deviations require explanation. Key Q: Is performance sustainable or transitory?*

Ratios & DuPont (S2)

ROE = NI/Avg.Eq **ROA** = NI/Avg.TA
DuPont: ROE = $\frac{NI}{Sales} \times \frac{Sales}{\text{Avg.TA}} \times \frac{\text{Avg.TA}}{\text{Avg.Eq}}$ = Margin × Turnover × Leverage
Profitability: Gross/EBITDA/EBIT/Net margins
Activity: AR/Inv/Payables/Fixed Asset Turnover
Leverage: D/TA, D/E Liquidity: Current, Quick, Int.Cov.
Valuation: P/E, P/S, P/B, TEV/EBITDA
NCI rule: use *total* NI+Equity *or* NI-to-common+Common Eq. Never mix.
Component isolation: assign one co.'s component to other's formula to quantify impact ("proforma recast").
▷ *Asset turnover has more permanent effect on ROA than margin (Soliman 2008).*

Piotroski F-Score (S2) — 9 binary signals

1. ROA > 0 2. CFO > 0 3. CFO > NI
2. ΔROA > 0 5. ΔGross Margin > 0
3. ΔAsset Turnover > 0
4. Δ(LTD/TA) < 0 (incl. curr. portion)
5. ΔCurrent Ratio > 0 9. No new equity to public

7–9 strong; **0–3** weak. ROA = NI (bef. disc.ops) / Beg.TA
▷ *Designed for low P/B neglected stocks. Works in portfolio (idiosyncratic shocks cancel); not a commodity/macro predictor. High F ⇒ profitability increases, positive earnings surprises, stock outperformance.*

Accruals & Earnings Quality (S4)

Total Accruals = $NI - CFO$
Accrual Ratio (CFA): $(NI - CFO - CFI)$/Avg.TA; also $(NI - CFO)$/Avg.TA
WC accruals: ΔCA − ΔCL (ex. cash/ST debt); volatile, short-term
Non-WC: D&A, def. taxes; stable, larger
Sign convention: CFS *backs out* accruals from NI ⇒ accrual = *opposite sign* of CFS line.
e.g. Inv. increase: (−98) on CFS ⇒ accrual = +98.
Non-cash test: IS charge = CFS add-back ⇒ 100% non-cash that period.
Inventory accrual ≠ ΔBS Inventory (acquisitions, write-offs, FX can differ).
CF manipulation: Tyco (investing → operating); Enron (investing → operating); WorldCom (operating → investing); Parmalat (falsified accounts).
▷ *Positive accruals ⇒ lower quality. Accr. reverse over time. Managers shift timing, not create profits.*

Days Metrics (S4–5)

Days Inv = 365 ÷ (COGS/Avg.Inv) Days AR = 365 ÷ (S/Avg.AR)
Days AP = 365 ÷ (COGS/Avg.AP) Days Unearn = 365 ÷ (S/Avg.Unearn)
▷ *Declining Days AR + booming sales = healthy. Rising Days Inv + backlog = ramp-up (not necessarily bad). Stable Days Unearn = mix stable; new high-days products offsetting low-days ones.*

Cookie Jar Reserves (S5–7)

Doubtful%: Allow/AR_g ($\text{AR}_g = \text{AR}_n$+Allow)
Returns/Warranty%: Addtn to Reserve/Sales
Dep.Life = Avg.Gross PPE/Dep.Exp (Gross = Net + Accum.); Rem.life = Net/Dep.Exp
LIFO→FIFO: ΔLIFO Res = COGS_L− COGS_F; FIFO COGS = LIFO COGS − ΔLIFO Res
LIFO layer liquidation: drawing down inv. levels forces old low-cost inventory into COGS ⇒ artificially *reduces* COGS ⇒ inflates profits. Firms must disclose this effect.
DTA Allowance: required when DTA "more likely than not" not realizable; ↑ allowance ⇒ ↑ tax expense.
▷ *Aggressive accruals foreshadow reversals. Accounting diffs can be legitimate; aggressive on one can be conservative on another.*

Special Items & Non-GAAP (S6–7)

Core/recurring: SG&A, R&D, royalties, amortization (same sign, every year).
Special: Restructuring, impairments, M&A costs, litigation, asset G/L, disc.ops.
Gray areas: Litigation (winding down vs. ongoing); Contingent consideration (recurring if M&A is core strategy, special if one-off). Justify with pattern + strategy.
Special items on CFS too: Debt extinguishment, investment G/L.
Adj. rule: Back out from NI; adjust tax; ignore BS impact. "Back out" ≠ "ignore."
Non-GAAP: Street earnings; excludes specials + sometimes recurring (stock comp). You decide if reasonable.
OCI (bypasses IS → equity): AFS sec., hedges, pension G/L, FX translation, IFRS reval.
▷ *Negative specials = cockroaches: more will follow. Non-recurring ≠ short-lived. Turnaround: specials ⇒ corrective actions ⇒ improvement eventually.*

Bankruptcy & Z-Scores (S3)

Z (public mfg): $1.2\frac{WC}{TA} + 1.4\frac{RE}{TA} + 3.3\frac{EBIT}{TA} + 0.6\frac{MVE}{BVL} + 1.0\frac{S}{TA}$
$Z > 3.0$ safe; 1.8–3.0 grey; $Z < 1.8$ distress
Z″ (any non-fin): $3.25 + 6.56\frac{WC}{TA} + 3.26\frac{RE}{TA} + 6.72\frac{EBIT}{TA} + 1.05\frac{BVE}{BVL}$
$Z'' > 5.85$ safe; 4.35–5.85 grey; $Z'' < 4.35$ distress. Maps to bond ratings.
EBIT = pretax income + int.exp. MVE = price × shares at FYE.
EBIT/TA and S/TA use *year-end* TA (not avg); other Z vars unspecified. *Z article cutoffs wrong; use slide values.*

Event	WC/TA	RE/TA	EBIT/TA	BVE/BVL
IPO/equity raise	↑	↑*	↑*	↑
Net loss	∘	↓	↓	↓
New debt	↑ST	∘	∘	↓
Asset write-off	∘	↓	↓	↓

▷ *IPO: CA↑, TA↑, BVE↑ (via APIC); EBIT and RE numerators unaffected. *For startup (RE<0, EBIT<0): ratios increase toward zero. For profitable co (RE>0, EBIT>0): RE/TA↓, EBIT/TA↓. Cash cushion ≠ fundamental health.*

Beneish M-Score (S8)

$$M = -4.84 + 0.92\,DSRI + 0.528\,GMI + 0.404\,AQI + 0.892\,SGI + 0.115\,DEPI - 0.172\,SGAI - 0.327\,LVGI + 4.679\,TATA$$

Var	Formula (num yr_t / denom yr_{t-1} unless noted)
DSRI	$(AR/S)_t/(AR/S)_{t-1}$
GMI	GM_{t-1}/GM_t
AQI	$(1-(CA+PPE_n)/TA)_t/(\cdots)_{t-1}$
SGI	S_t/S_{t-1}
DEPI	$(D/(D+PPE_n))_{t-1}/(\cdots)_t$
SGAI	$(SGA/S)_t/(SGA/S)_{t-1}$
LVGI	$((LTD+CL)/TA)_t/(\cdots)_{t-1}$
TATA	$(NI_{\text{bef.disc}} - CFO)/TA_t$

Higher M = more risk. 2+ vars near manipulator col = red flag.
Set AQI= 1 if $(CA+PPE) = TA$ in $t-1$; PPE incl. capit. leases.
Common errors: AQI: use net PPE (not bldgs); CA+PPE / TA (not CL). DEPI: depreciation only (not D&A). LVGI: LTD + CL in numerator / TA. TATA: divide by TA.
Why errors cancel: ratio-of-ratios ⇒ num/denom errors offset; TATA small ⇒ big rel. error, small abs. effect; LVGI negative coeff. moderates.
▷ *Intersegment transactions net to zero in consol. FS ⇒ M-Score cannot detect segment-level manipulation.*

Dechow F-Score (S8)

$$\hat{y} = -7.893 + 0.790r + 2.518c_r + 1.191c_i + 1.979s + 0.171c_s - 0.932c_\rho + 1.029\,\text{iss}$$

Prob $= e^{\hat{y}}/(1+e^{\hat{y}})$; F = Prob/0.0037

F> 1 above normal; F> 1.85 substantial; F> 2.45 high

Var	Definition
r (rsst_acc)	ΔNon-cash NOA/Avg.TA
c_r (ch_rec)	ΔAR_n/Avg.TA
c_i (ch_inv)	ΔInv/Avg.TA
s (soft_assets)	$(TA - PPE_n - Cash)/TA$
c_s (ch_cs)	$\%\Delta(S - \Delta AR)$
c_ρ (ch_roa)	$\Delta(NI/\text{Avg.TA})$
iss (issue)	= 1 if LTD or equity issued

Non-cash NOA = Equity − Pref.Stk − Cash. NI bef. disc.ops; PPE incl. cap. leases; equity incl. SBC.

▷ *Score mainly driven by soft_assets and issue. Asset-light (low D&A) ⇒ CFO < NI is not automatically a bad signal (D&A is usually the biggest accrual).*

Benford's Law (S8)

Distribution of the occurrence of 1st digit of numbers in a group of numbers. Tends towards lower numbers.

$P(d) = \log_{10}(1 + 1/d)$

Applies to naturally occurring data (revenues, populations). Does NOT apply to assigned numbers (zip codes, phone numbers). Deviations = manipulation signal.

▷ *Fraudsters can't generate convincingly random-looking numbers.*

Valuation Multiples (S2)

Trailing P/E = P/TTM EPS; Fwd P/E = P/NTM EPS; PEG = (P/E)/LT growth

MVE/Pretax Income: avoids tax distortions.

TEV = MVE + LTD + Curr.LTD − Cash; TEV/EBITDA or TEV/EBIT.

Consistency: entity-level numerator (TEV) ↔ pre-debt denominator (EBIT/EBITDA); equity-level (MVE) ↔ post-debt (NI, EPS, pretax).

▷ *Special items distort: transitory vs. recurring? Single-peer caveat: more comps = stronger conclusion. Watch units (millions vs. thousands).*

Segments & NCI (S9)

Reportable segment if any: rev ≥ 10%, profit ≥ 10%, or assets ≥ 10% of all segs. Combined reported segs ≥ 75% of consol. revenue.

Required disclosures: P&L per seg; often rev/assets/capex/dep; also product/geo/customer.

Intersegment transactions eliminated in consolidation.

NCI(IS) = minority share of sub NI; NCI(BS) = minority share of sub net assets (in equity).

Redeemable NCI → liabilities or temp. equity.

▷ *Check: Total Equity − NCI → NI to parent. NCI = "a segment the company doesn't own." Treat NI and equity consistently.*

Forecasting Revenue & Expenses (S10–11)

Revenue = P × Q. Growth mean-reverts to **3–7%**; typical 7–11%; wide variance.

Revenue recognition (5-step, FASB/IASB eff. Jan 2018): (1) Identify contract; (2) Identify performance obligations; (3) Determine transaction price; (4) Allocate price to obligations; (5) Recognize when/as each obligation satisfied.

Tricky for: subscriptions, bundles, commissions, licenses, long-term contracts.

Expenses: %Sales or regression: Exp $= a + b \cdot S$. Adj for scale (↓), lumpiness (↑ST), sticky costs (↑ST in downturns).

Dep.Exp ≈ (1/Dep.Life) × Avg.Gross PPE; apply to existing PPE + capex.

▷ *Revenue is the engine; all forecasts follow from it. FS articulate: IS → RE → CFS. Higher fixed costs ⇒ more margin volatility.*

Stock Compensation (S11)

Fair value expensed as earned (options: Black-Scholes; RSUs: price at grant). Real cost.

IS: salary exp. CFS: add back as non-cash op. adj. (like D&A).

BS: APIC ↑, DTA, treasury stock.

Often excluded from Non-GAAP; Buffett/Damodaran: real operating expense; dilutes shareholders.

Key Conceptual Rules (Quiz Focus)

Interpreting questions:

- State formula → apply it → acknowledge alternative interpretations → conclude
- "Small accrual" can mean demand met *or* supply constrained — say both
- If IS charge = CFS add-back ⇒ 100% non-cash; state why (e.g. prior-period deposit)
- Link: accounting signal → economic logic → forward-looking implication

F/Z model philosophy: pre-defined recipes check analyst bias; component analysis yields same insight as DuPont drill-down.

Benchmarking: accounting diffs can be legitimate; aggressive on one ⇔ conservative on another; be consistent when adjusting benchmarks; make reasonable assumptions when data is missing.

Piotroski: high score ⇒ profitability ↑, positive surprises, stock outperformance; portfolio effect ≠ single stock guarantee.

M-Score gray zone (−2.22 to −1.78): inconclusive — could signal other issues even if specific manipulation not detected.

Z" gray zone (4.35–5.85): monitor closely; one bad year can push into distress.

Acronym Glossary

Financial Statements & Locations

Acr.	Meaning	Where
NI	Net Income	IS (bottom line)
EPS	Earnings Per Share	IS (below NI) or calc
EBIT	Earnings Before Interest & Tax	Calc: pretax + int.exp.
EBITDA	EBIT + Dep. + Amort.	Calc (not on stmt)
COGS	Cost of Goods Sold	IS
GM	Gross Margin (= Sales − COGS)	IS
SG&A	Selling, General & Admin. Exp.	IS
R&D	Research & Development Exp.	IS
D&A	Depreciation & Amortization	CFS (op. add-back) / FN
SBC	Stock-Based Compensation	IS (various lines) + CFS
OCI	Other Comprehensive Income	Stmt of Comp. Inc. / Eq.
CFO	Cash Flow from Operations	CFS (operating section)
CFI	Cash Flow from Investing	CFS (investing section)
CFF	Cash Flow from Financing	CFS (financing section)
TA	Total Assets	BS
CA	Current Assets	BS
CL	Current Liabilities	BS
WC	Working Capital (CA − CL)	BS (calc)
AR	Accounts Receivable	BS (current assets)
Inv	Inventory	BS (current assets)
AP	Accounts Payable	BS (current liabilities)
PPE	Property, Plant & Equipment	BS (non-current assets)
LTD	Long-Term Debt	BS (long-term liabilities)
RE	Retained Earnings	BS (equity section)
BVE	Book Value of Equity	BS (equity section)
BVL	Book Value of Liabilities	BS (total liabilities)
MVE	Market Value of Equity	Calc: price × shares
TEV	Total Enterprise Value	Calc: MVE+LTD+CurrLTD−C
NCI	Noncontrolling Interest	IS (below NI) + BS (equity)
APIC	Additional Paid-In Capital	BS (equity section)
DTA	Deferred Tax Asset	BS (assets)
LIFO	Last In, First Out (inventory)	FN (inventory note)
FIFO	First In, First Out (inventory)	FN (inventory note)

Model Variables

Acr.	Meaning
DSRI	Days Sales in Receivables Index (M)
GMI	Gross Margin Index (M)
AQI	Asset Quality Index (M)
SGI	Sales Growth Index (M)
DEPI	Depreciation Index (M)
SGAI	SG&A Expenses Index (M)
LVGI	Leverage Index (M)
TATA	Total Accruals to Total Assets (M)
NOA	Net Operating Assets (F-Score)
rsst_acc	ΔNon-cash NOA / Avg.TA (F)
ch_rec	ΔReceivables / Avg.TA (F)
ch_inv	ΔInventory / Avg.TA (F)
soft_assets	(TA−PPE−Cash)/TA (F)
ch_cs	% Δ Cash Sales (F)
ch_roa	Δ Return on Assets (F)
iss	Debt or equity issuance (F)

Standards & Regulatory

Acr.	Meaning
GAAP	Generally Accepted Accounting Principles
IFRS	International Financial Reporting Standards
FASB	Financial Accounting Standards Board (US)
IASB	International Accounting Standards Board
SEC	Securities & Exchange Commission
10-K	Annual report (IS, BS, CFS, FN in Item 8)
MD&A	Management Discussion & Analysis (Item 7)
AAER	SEC Accounting & Auditing Enforcement Release
ROE	Return on Equity (calc from IS + BS)
ROA	Return on Assets (calc from IS + BS)
P/E	Price-to-Earnings (mkt price / EPS)
PEG	Price/Earnings-to-Growth